UMMEHANI BURHANUDDIN

Software Engineer (AI/ML) · Computer Science · 2027 Graduate

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE (AI/ML) undergraduate graduating in 2027, with strong foundations in Data Structures, Algorithms, and Object-Oriented Programming (200+ LeetCode problems solved). Builds distributed, service-oriented systems — multi-agent AI pipelines, real-time streaming architectures, and RAG systems — with an emphasis on scalability, fault tolerance, and explicit, defensible design trade-offs over black-box AI usage. Comfortable across Python, Java, FastAPI, React, and cloud fundamentals, with hands-on Docker and AWS (EC2, S3) deployment experience.

EDUCATION

B.Tech – Computer Science Engineering (AI & ML) | Expected Graduation: 2027

Lakshmi Narain College of Technology, Bhopal

Relevant Coursework: Data Structures & Algorithms · Object-Oriented Programming · Machine Learning · Deep Learning · NLP · DBMS · Operating Systems · Computer Networks · Computer Vision

TECHNICAL SKILLS

Languages: Java, Python, SQL, C++, Unix/Bash Scripting

Systems & Backend: Distributed Systems, Microservices, REST APIs, WebSockets, Streaming Architecture, Flask, FastAPI

AI / ML: Machine Learning, Deep Learning, NLP, Generative AI, RAG, Multi-Agent AI, Agentic Workflows, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, LLMs, Whisper, LangChain

Frontend: React, Component-based UI, API Integration

Cloud & DevOps: Docker, AWS (EC2, S3), Deployment Fundamentals, Linux, Git/GitHub version control

Databases: MySQL, MongoDB, PostgreSQL

PROJECTS

Quantum AI Research Agent — *FastAPI · React · Python · Pandas · yFinance · Multi-Agent AI* [GitHub]

- Architected a 4-agent, service-oriented financial research system (Technical, Fundamental, Sentiment, and a Master Synthesis agent) after an initial single-LLM-prompt approach produced inconsistent, hard-to-interpret outputs — decomposing into independent agents improved explainability and made the system easier to debug, test, and extend in isolation.
- Used a consensus-based synthesis design instead of fixed weighting: each specialized agent independently produces a confidence-scored analysis, which the Master Agent merges into a final Buy/Hold/Sell recommendation with an overall confidence score.
- Integrated 3 live external data sources (Yahoo Finance, News API, company fundamentals) across 5 timeframes (15m to 1W) and implemented API response caching to cut redundant upstream calls and keep the system responsive under concurrent requests.
- Built a full-stack platform (React + FastAPI) to surface structured, human-interpretable investment insights in real time.

Real-Time AI Meeting Assistant — *Whisper · FAISS · LangChain · WebSockets · FastAPI · React* [GitHub]

- Designed a real-time, WebSocket-based streaming architecture — replacing HTTP polling — to push live transcription updates to clients, cutting redundant client requests and reducing perceived latency under concurrent sessions.
- Built a live transcription and summarization pipeline using Whisper Base, chosen over larger Whisper variants for the accuracy/latency trade-off real-time meetings require.
- Designed a FAISS-backed semantic memory layer for cross-session meeting retrieval, choosing FAISS over a managed vector DB for lightweight, self-hosted integration and lower operational overhead.
- Delivered an end-to-end, fault-tolerant pipeline — audio input, transcription, speaker segmentation, LLM-based summarization, action-item extraction, and report generation — orchestrated with LangChain.

KAKSHA — AI-Powered Personalized Learning Path Recommender — *FastAPI · Next.js 16 (React 19, TypeScript) · Groq · Sentence-Transformers · PyTorch · Supabase (pgvector) · HCL Tech Hackathon* [GitHub]

- Built for the HCL Tech Hackathon to address a real gap in online learning: generic course catalogs give every learner the same list regardless of what they know, causing redundant content and disengagement — KAKSHA instead parses free-text background (e.g. "I know basic Python syntax, but I've never built APIs or used databases") to pinpoint exact skill gaps.
- Used Groq for high-speed LLM profile extraction and personalized explanations, paired with a locally-run Sentence-Transformers model (all-MiniLM-L6-v2) via PyTorch for embeddings — keeping embedding cost and latency off the LLM path while reserving Groq for reasoning-heavy steps.
- Designed a Supabase (PostgreSQL + pgvector) retrieval layer that matches each identified skill gap to the most relevant learning resources by cosine similarity, generating a customized, week-by-week roadmap sequenced to the learner's actual starting point.
- Built a full-stack application with a FastAPI (Python) backend and a Next.js 16 (React 19, TypeScript) frontend styled with Tailwind CSS and Lucide React.

CERTIFICATIONS

Introduction to Machine Learning — NPTEL (IIT Madras) · Programming in Java — NPTEL (IIT Kharagpur) · Python Essentials 1 — Cisco Networking Academy (2025)

ACHIEVEMENTS & LEADERSHIP

- Top 100 finisher, Capgemini BuildAThon; Top 50 finisher, Capgemini Side Quest.
- Solved 200+ problems on LeetCode with consistent competitive programming practice, focused on algorithmic problem solving and optimization.
- Smart India Hackathon Participant — developed AI/ML solutions for national-scale challenges.
- Active member of Orators Club, building technical presentation and stakeholder communication skills.